Programming and Spreadsheet Modeling

Mathematical Model Building

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Management Science Models
Mathematical Model Elements

- Assumed values:
  - Parameters
- Dependent variables:
  - Takes on the value of what we're trying to model
- Independent variables:
  - Takes on the values of those things which impact the value of the independent variable
- Mathematical relationship:
  - Function relating the parameters, independent and dependent variables

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Management Science Models
Variables and functions

- The perimeter of a rectangle equals twice the width plus twice the height.

perimeter = 2 * width + 2 * height

- A function can also be written showing the input variables.

$$perimeter = f(width, height)$$

- Width and height are independent variables which can be varied. They can also be identified by letters.

$$X_1 = width$$

$$X_2 = height$$

- Perimeter is a dependent variable, as its value depends on the values selected for width and height, again written as a letter.

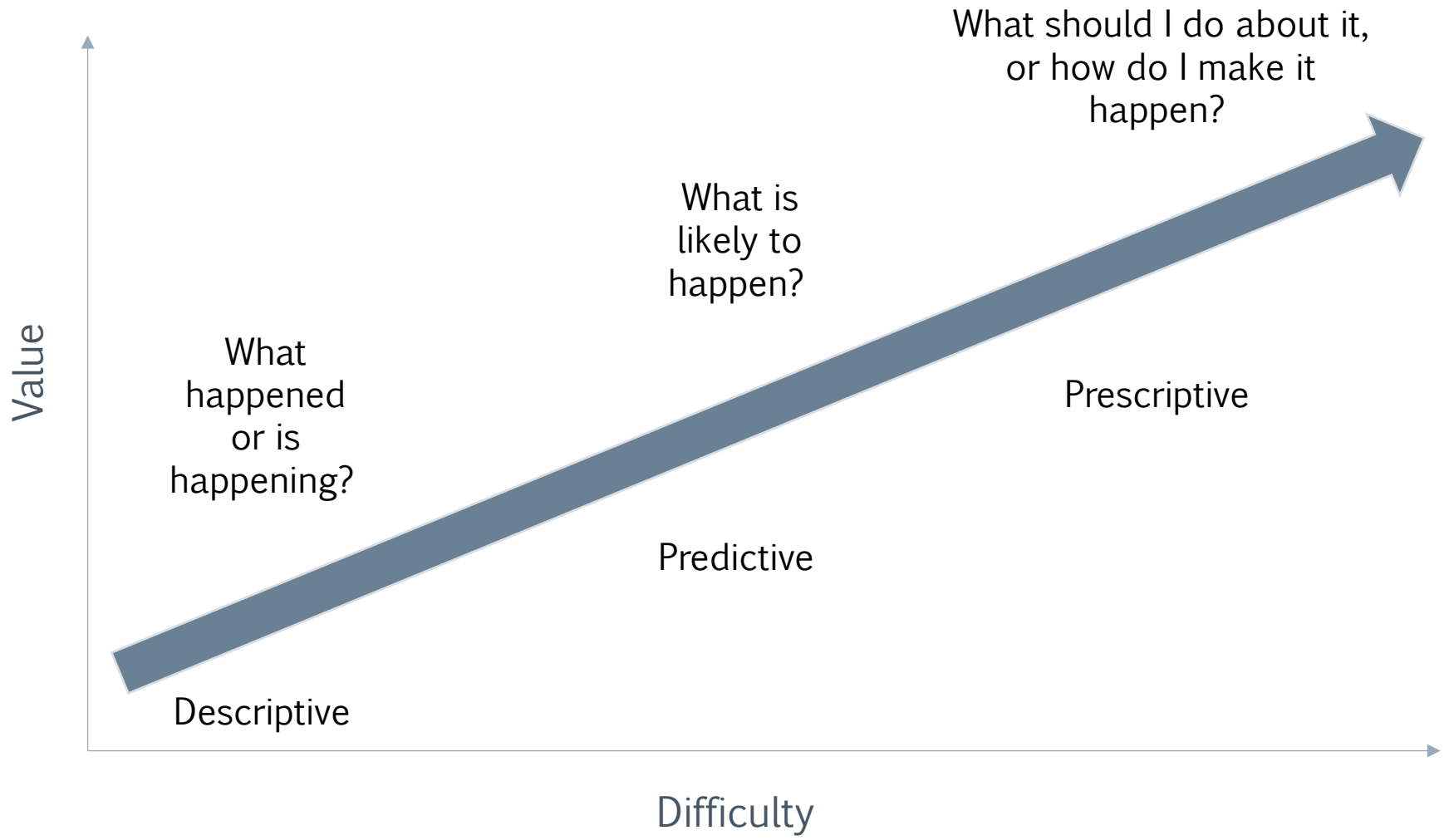
$$Y = perimeter$$

- A function then is the relationship between the independent and dependent variables

$$Y = f(X_1, X_2)$$

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Business Analytics Modeling Progression



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Management Science Models
Model Categorization Example - Descriptive

- **Situation**
  - A store manager wants to determine how long customers will wait for checkout.
- **Independent variable values (X)**: **unknown**
  - Customer arrival rate (λ) and service rate (μ)
- **Dependent variable values (Y):** **what we want to know**
  - Expected customer waiting time
- **Function / relationship:** **known**
  - $W_q = f(\lambda, \mu)$ from queuing theory
- **What we know?**
  - We know the mathematical relationship between arrival rate, service rate, and waiting time, but actual arrival/service rates are uncertain.
- **Model Purpose**
  - **Describe what happens** to values of Y given uncertain operating conditions

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Management Science Models
Model Categorization Example - Predictive

- **Situation**
  - A retailer wants to predict monthly sales based on advertising spending.
- **Independent variable values (X)**: **known** or **under control of decision maker**
  - Advertising spending
- **Dependent variable values (Y):** **known** from past data
  - Monthly sales
- **Function / relationship:** **unknown or assumed**
  - Linear regression: $Sales = \beta_0 + \beta_1(Advertising) + \epsilon$
- **What is known?**
  - We can choose advertising spending, but **do not know** the relationship between that advertising spend and sales, even if we may know the form of the relationship.
- **Model Purpose**
  - **Discover/estimate the relationship** (coefficients) so that we can predict Y

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Management Science Models
Model Categorization Example - Prescriptive

- **Situation**
  - A manufacturer wants to determine how many units to produce to maximize profit.
- **Independent variable values (X)**: **under control of decision maker**
  - Units of Product A and Product B to produce
- **Dependent variable values (Y):** **unknown**
  - Total profit
- **Function / relationship:** **known**
  - $Profit = \$40A + \$30B$, subject to resource constraints
- **What is known?**
  - We know the profit contribution and resource requirements for each product.
- **Model Purpose**
  - **Choose the values for X** to obtain the best possible value of Y

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Management Science Models
Model Categorizations

- Descriptive:
  - *I know the function, but I don't know what Y will be for different Xs.*
  - ***Describes*** *what happens to Y for values of X*

- Predictive:
  - *I know past Xs and Ys, but I don't know the function connecting X to Y.*
  - *Finding the relationship enables* ***predicting*** *Y for any X*

- Prescriptive:
  - *I know the function, and I choose X to get the Y that I want.*
  - ***Prescribes*** *the values of X to obtain the best value of Y*

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